

Chapter 05 Homework — Tests for irreducibility

Mikey Ferguson

September 22, 2026

Problem 05.4. *Suppose that K is a field and that f and g are relatively prime in $K[x]$. Show that $f - yg$ is irreducible in $K(y)[x]$.*

Problem 05.6. *Suppose that R is an integral domain and that*

$$f = f_0 + f_1x + \cdots + f_nx^n \in R[x]$$

has the property that $f_0, \dots, f_n$ are relatively prime. Suppose that p is a prime in R , and that $p \mid f_i$ for $1 \leq i \leq n$, while p does not divide f_0 and p^2 does not divide f_n . Show that f is irreducible in $R[x]$.

Problem 05.7. *Show that if p is a prime number then $x^n - p$ is irreducible in $\mathbb{Q}[x]$.*

Problem 05.8. *Let A denote the field of real numbers which are algebraic over $\mathbb{Q}$. Show that $[A : \mathbb{Q}] = \infty$.*

Problem 05.10. *Show that $x^5 - 4x + 2$ and $x^4 - 4x + 2$ are irreducible over $\mathbb{Q}(i)$.*